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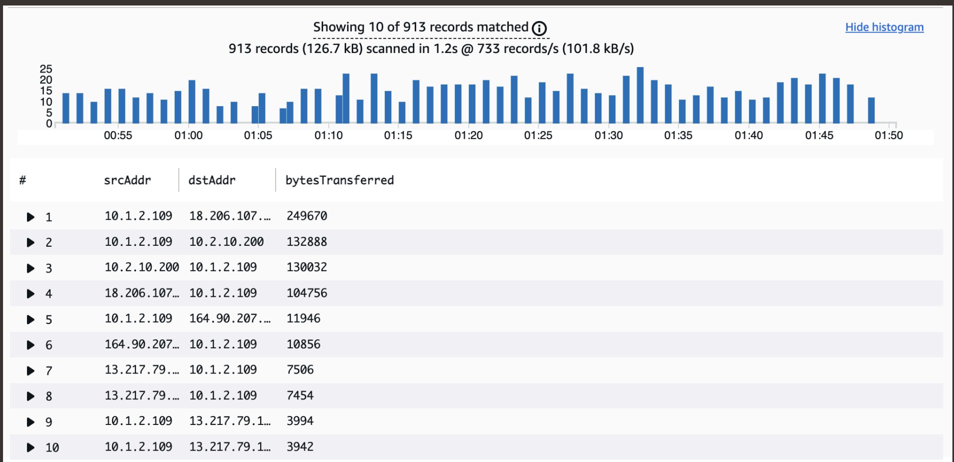
VPC Monitoring with Flow Logs

CH Arpita Chowdhury



#	srcAddr	dstAddr	bytesTransferred
1	10.1.2.109	18.206.107.…	249670
2	10.1.2.109	10.2.10.200	132888
3	10.2.10.200	10.1.2.109	130032
4	18.206.107…	10.1.2.109	104756
5	10.1.2.109	164.90.207.…	11946
6	164.90.207…	10.1.2.109	10856
7	13.217.79.…	10.1.2.109	7506
8	13.217.79.…	10.1.2.109	7454
9	10.1.2.109	13.217.79.1…	3994
10	10.1.2.109	13.217.79.1…	3942

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private cloud that provides an isolated and customizable network environment in AWS where you can deploy resources securely. It is useful because it gives you granular control over your networking—such as IP ranges, subnets, routing, and firewall rules—allowing you to build secure, scalable, and compliant cloud architectures.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to monitor VPC with flowlogs.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was seeing all the log flow events and seeing the top query transfer in a chart form.

This project took me...

This project took me 2 hours.

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Arpita Chowdhury
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In the first part of my project...

Step 1 - Set up VPCs

In this step, I will set up two VPCs because I will use this VPCs to complete this project.

Step 2 - Launch EC2 instances

In this step, I will launch EC2 instances in each VPCs because I will use them to test VPC peering connection later.

Step 3 - Set up Logs

In this step, I will set up a way to track all inbound and outbound network traffic and a space that stores all of these records because I will use them to set up flow logs.

Step 4 - Set IAM permissions for Logs

In this step, I will give VPC Flow Logs the permission to write logs and send them to CloudWatch and finish setting up subnet's flow log because this will set up a flow log IAM policy and role.

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Arpita Chowdhury
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Multi-VPC Architecture

I started my project by launching 2 VPCs and 1 public and 0 private subnet.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique because VPC must have a unique IPv4 CIDR block so the IP addresses of their resources don't overlap. Having overlapping IP addresses could cause routing conflicts and connectivity issues.

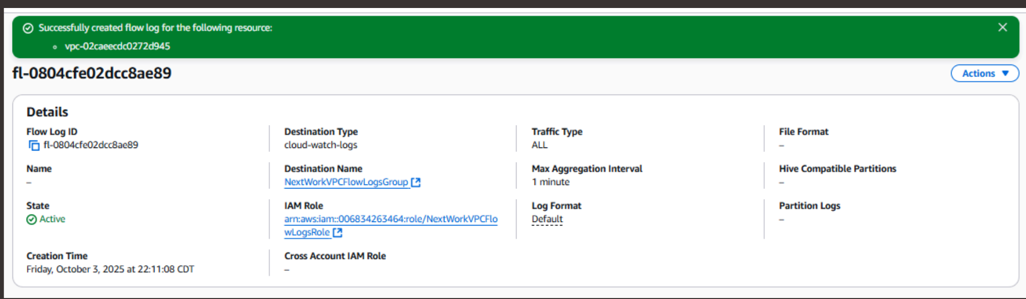
I also launched EC2 instances in each subnet

My EC2 instances' security groups allow in ICMP traffic from all IP addresses, not just from the other VPC. This is because I will be doing a new ping test.

Logs

Logs are data files that record events, activities, and operations from AWS services, applications, and infrastructure. They provide detailed information that is used for monitoring, troubleshooting, security analysis, and auditing.

Log groups are collections of log streams that contain log events from your applications and AWS services. They serve as a way to organize logs by defining shared settings for retention, access control, and monitoring for all the log streams within them.

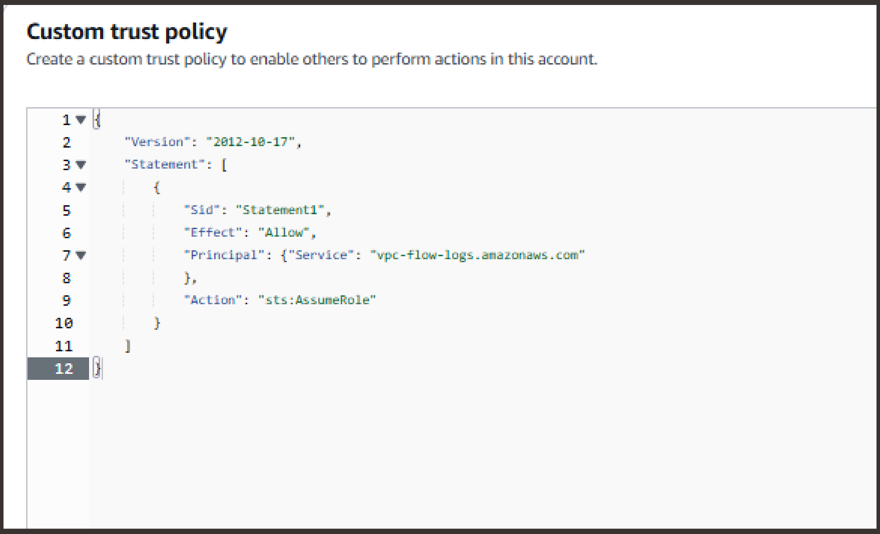


IAM Policy and Roles

I created an IAM policy because VPC flow logs by default don't have the permission to record logs and store them in CloudWatch log group. This policy makes sure that VPC can now send log data to log group.

I also created an IAM role because if I bundle IAM policies together, I have to create an IAM role. Then I assign this role to AWS services or users, giving them the permissions included in the attached policies.

A custom trust policy is specific type of policy! They're different from IAM policies. While IAM policies help you define the actions a user/service can or cannot do, custom trust policies are used to very narrowly define who can use a role.



Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "Statement1",
            "Effect": "Allow",
            "Principal": {"Service": "vpc-flow-logs.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step, I will get instance 1 to send message to instance 2 because I want to test VPC peering.

Step 6 - Set up a peering connection

In this step, I will set up a connection link between VPCs because I want to create a peering connection and configure Route Tables.

Step 7 - Analyze flow logs

In this step, I will analyze flow logs because I want to review the flow logs recorded about VPC 1's public subnet and analyze the flow logs to get insight.

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Arpita Chowdhury
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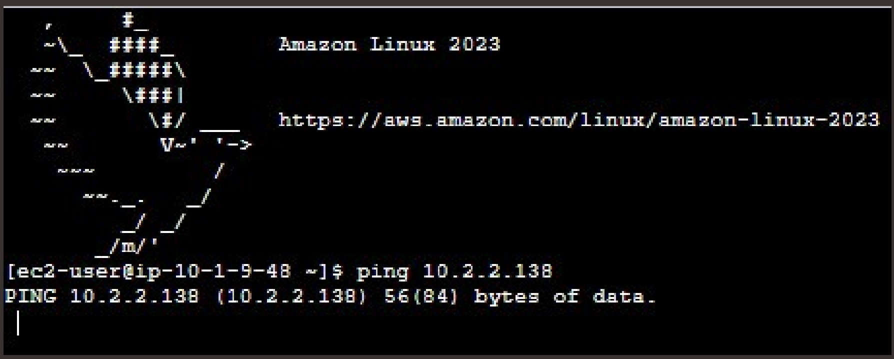
Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means VPC 2's network might be blocking the type of traffic used with ping i.e. ICMP (Internet Control Message Protocol) traffic.



I could receive ping replies if I ran the ping test using the other instance's public IP address, which means Instance 2 is correctly configured to respond to ping requests, and Instance 1 can actually communicate with Instance 2 if it traffic goes across the public internet.

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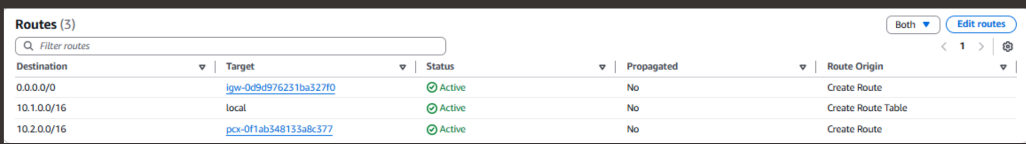
Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because VPC 1’s route table doesn’t have a route to Instance 2’s subnet/CIDR via the correct attachment (peering/TGW/VPN). Packets to Instance 2’s private IP don’t match any prefix except local (VPC 1 only) and get dropped so the ping fails

To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that I can make sure traffic from VPC 1 can go to VPC 2 route table helps with the traffic direction.



Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0d9d976231ba327f0	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	pcx-0f1ab348133a8c377	Active	No	Create Route

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Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means the connectivity issue between VPC 1 and 2 is resolved by setting up the peering architecture.

```
Last login: Fri Oct 31 23:02:45 2025 from 18.206.107.27
[ec2-user@ip-10-1-9-48 ~]$ ping 10.1.9.48
PING 10.1.9.48 (10.1.9.48) 56(84) bytes of data.
64 bytes from 10.1.9.48: icmp_seq=1 ttl=127 time=0.024 ms
64 bytes from 10.1.9.48: icmp_seq=2 ttl=127 time=0.032 ms
64 bytes from 10.1.9.48: icmp_seq=3 ttl=127 time=0.032 ms
64 bytes from 10.1.9.48: icmp_seq=4 ttl=127 time=0.031 ms
64 bytes from 10.1.9.48: icmp_seq=5 ttl=127 time=0.033 ms
64 bytes from 10.1.9.48: icmp_seq=6 ttl=127 time=0.032 ms
64 bytes from 10.1.9.48: icmp_seq=7 ttl=127 time=0.039 ms
64 bytes from 10.1.9.48: icmp_seq=8 ttl=127 time=0.036 ms
64 bytes from 10.1.9.48: icmp_seq=9 ttl=127 time=0.029 ms
64 bytes from 10.1.9.48: icmp_seq=10 ttl=127 time=0.031 ms
64 bytes from 10.1.9.48: icmp_seq=11 ttl=127 time=0.031 ms
64 bytes from 10.1.9.48: icmp_seq=12 ttl=127 time=0.034 ms
64 bytes from 10.1.9.48: icmp_seq=13 ttl=127 time=0.031 ms
64 bytes from 10.1.9.48: icmp_seq=14 ttl=127 time=0.032 ms
64 bytes from 10.1.9.48: icmp_seq=15 ttl=127 time=0.035 ms
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64 bytes from 10.1.9.48: icmp_seq=18 ttl=127 time=0.034 ms
64 bytes from 10.1.9.48: icmp_seq=19 ttl=127 time=0.032 ms
64 bytes from 10.1.9.48: icmp_seq=20 ttl=127 time=0.030 ms
```

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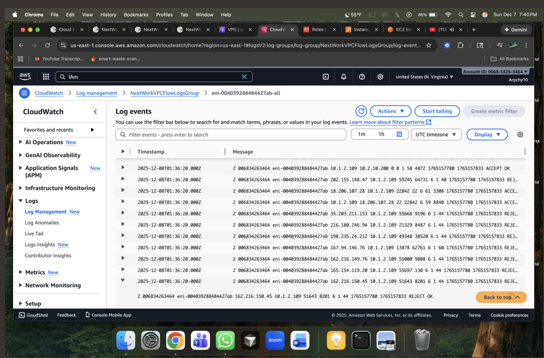
Analyzing flow logs

Flow logs tell us about who communicated with whom, when they communicated, how they communicated (protocols/ports), whether the traffic was allowed or denied, and how much data was transferred.

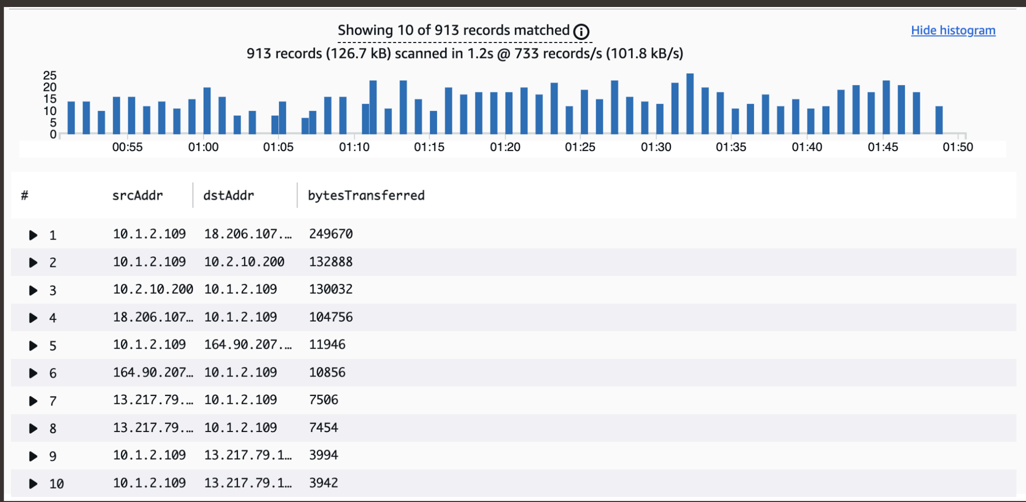
For example, the flow log I've captured tells us the details of each network connection involving my ENI (Elastic Network Interface) including the source and destination IP addresses, ports, protocol used (TCP/UDP), how many bytes were transferred, whether the traffic was accepted or rejected, and the security decision taken for each flow.



Logs Insights

Logs Insights is a fully managed, scalable log analytics tool within CloudWatch that allows you to run fast queries on huge volumes of log data, filter and extract fields, create visualizations, and build dashboards all without needing to manage any servers or storage.

I ran the query Top 10 byte transfers by source and destination IP addresses. This query analyzes the VPC Flow Logs to identify the top network flows based on the amount of data transferred. It groups the log records by source and destination IP addresses, sums the total bytes exchanged between each pair, and then returns the flows with the highest volumes.



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